

English Composition Image Automatic Scoring Based on Multi-modal Large Language Models

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Abstract

Large language models (LLMs) bring significant opportunities in the field of education evaluation. Nowadays, most existing LLMs are introduced to generate reasonable scores for electronic and structured documents. However, such systems require huge efforts for manual typewriting or advanced optical character recognition techniques. To this end, this work directly processes the scanning copy of English composition with multi-modal LLMs. Specifically, this research aims to utilize multi-modal LLMs for automated essay scoring (AES) and evaluate its reliability and accuracy. We take the English composition images of 1,511 tenth-grade examinees in a standardized test environment as the research objects for automated scoring, and explore the performance of the multi-modal LLM GPT-4o in evaluating composition image data. The study compares the consistency of the composition image scores of GPT-4o under zero-shot and two-shot prompts with the scores of the marking experts, and verifies them through multiple indicators such as the exact agreement coefficient, Pearson correlation coefficient, Coefficient of determination, Root Mean Square Error, and the probability that the GPT-4o score falls within a given confidence interval. Especially, comparing the scores given by experts, the R-square of the GPT-4o scoring reaches 0.66. The results show that GPT-4o has the ability to learn from two samples through prompt, can quickly adapt and provide high-quality and personalized evaluations for examinees, and can provide valuable support for human scoring.

CCS Concepts

• **Social and professional topics** → Professional topics; Computing education; Student assessment.

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Keywords

Multi-modal large language model, GPT-4o, Automated Essay Scoring, Prompt Learning

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1 Introduction

In October 2020, the Central Committee of the Communist Party of China and the State Council issued the "Overall Plan for Deepening the Reform of Educational Evaluation in the New Era"^[1]. In March 2024, Minister Huai Jinpeng of the Ministry of Education pointed out during the National People's Congress and Chinese People's Political Consultative Conference^[2]. The deep integration of educational evaluation and artificial intelligence is becoming the main theme of educational evaluation reform in the new era^[3].

In the evaluation of English subjects, writing is a core link that comprehensively examines examinees' English proficiency and can deeply demonstrate students' proficiency and mastery level of English vocabulary, grammar, and other aspects^[4]. At present, most English subject examinations in our country still adopt the traditional paper-and-pencil examination method. The educational evaluation authorities convert examinees' paper answer sheets into electronic image documents through high-speed scanners. Then, organizing marking experts to score the segmented composition images. However, this traditional manual scoring process is on the one hand influenced by the subjective emotions of marking experts; on the other hand, due to the heavy evaluation tasks, it is difficult to provide personalized evaluation and feedback for each candidate. Therefore, it is facing challenges in efficiency improvement and technological innovation^[5].

Automated Essay Scoring (AES) offers the possibility to address these challenges in the field of English composition evaluation. AES involves the use of computer algorithms to evaluate and provide feedback on student essays^[6]. AES helps standardize the evaluation process, reduce subjective factors, and ensure the consistency

and fairness of evaluation. The automatic essay scoring system can also greatly reduce the burden of evaluation work and free up more time for classroom teaching and individual guidance^[7]. In 1966, Ellis Page developed the world's first automatic essay scoring system, Project Essay Grade (PEG)^[8]. It scores compositions by analyzing their surface features and content features. However, it ignores information such as word order, syntax, and logic and needs to be used in combination with variables reflecting language forms. Intelligent Essay Assessor (IEA)^[9] uses Latent semantic analysis technology to transform text into vector representations in a high-dimensional semantic space and evaluate composition quality by calculating the semantic similarity between compositions and standard model essays or specific topics. The scoring method of Electronic Essay Rater (E-rater)^[10] is based on a linear regression model and mainly uses toolkits based on Microsoft natural language processing and corpus-based methods for modeling to analyze articles. In 1999, the Educational Testing Service in the United States used the E-rater system to score GMAT compositions, and in 2005, it was also used in the TOEFL test^[11]. However, the scoring results of Intelligent Essay Assessor and E-rater highly depend on the quality and diversity of training data. If there are biases or limitations in the training data, inaccurate situations may occur when scoring other types of compositions. The pioneering use of CNN and RNN in Automated Essay Scoring systems represents a major step forward in educational assessment. In 2018, a significant milestone in the evolution of natural language processing was marked with the introduction of the ELMo (Embeddings from Language Models) model by Peters et al^[12]. ELMo pre-trains models through large-scale corpora and fine-tunes them for specific tasks. Therefore, it highly depends on a large amount of high-quality labeled data for training. In addition, its scoring decision-making process is difficult to explain and is usually considered a "black box."

Reviewing numerous domestic and foreign research and exploration practices, existing automatic essay scoring systems face multiple limitations. First, the current research objects mainly focus on the electronic text content of compositions and only analyze and process the text itself. However, converting examinees' compositions into text is a labor-intensive task. Moreover, this conversion will lose the information about whether the examinees' test papers are neatly written, that is, the score for neat handwriting on the test paper cannot be obtained. Second, automatic essay scoring systems rely on complex machine learning models, large data sets, and highly optimized algorithms, and the development and maintenance require professional technical knowledge and skills. Third, such systems usually require a large amount of corpus data sets for training to adapt to various English writing tasks of different themes and types.

Since the release of ChatGPT by OpenAI in late November 2022, the accessibility of GPT technology has been significantly enhanced. This development has led to suggestions that ChatGPT could be utilized for AES applications^[13]. In May 2024, OpenAI released their multimodal LLM, GPT-4o, which is a web-based platform that is available to virtually everyone. Supported by GPT-4o's large amount of multimodal training data and continuous updates, this paper delves into the feasibility and prospects of using GPT-4o to directly conduct automatic essay scoring on composition image

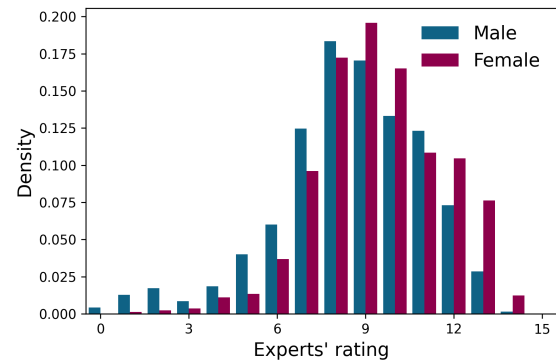


Figure 1: Score distribution of male and female examinees under experts' ratings.

data, with a particular focus on exploring its performance in terms of accuracy and reliability. We examine the potential role linguistic features could play in improving the accuracy of AES using GPT-4o.

2 Methods

2.1 Dataset

In this study, we took the images of 1,511 real English compositions of tenth - grade examinees in an administrative district of Shanghai as the research objects. These 1,511 examinees participated in the unified examination in 2024 under the unified examination time and environment, thus ensuring the fairness and validity of the response content. The educational authority of this administrative district organized senior marking experts to assign a score ranging from 0 to 15 to each composition response according to the unified scoring standard. The scoring adopted a double - evaluation - plus - arbitration method to ensure the authority and reliability of the scores. From the score data given by the marking experts, the minimum score is 0 points, the average score of the compositions is 9.21, the median is 9, the standard deviation is 2.34, and the maximum value is 15 and only one person got the full score of 15. In addition, the score distribution of examinees of different genders is shown in Figure 1. It can be seen from the figure that the number of female examinees in the high - score segment is greater than that of male examinees. These data provide an important basis and reference for further in - depth analysis, helping us better understand the overall situation of the research objects and the performance differences of different genders in relevant indicators.

2.2 GPT-4o

GPT-4o is an advanced multi-modal large language model based on the Transformer architecture. Compared with other language models, the training data of GPT-4o is multi-sourced and dynamic. It can not only process text information but also integrate and analyze data of multiple modalities such as images and audio. The data sources include a large amount of publicly available Internet data, data obtained through cooperation with publishers, and specifically annotated data. Moreover, these data are continuously updated. In

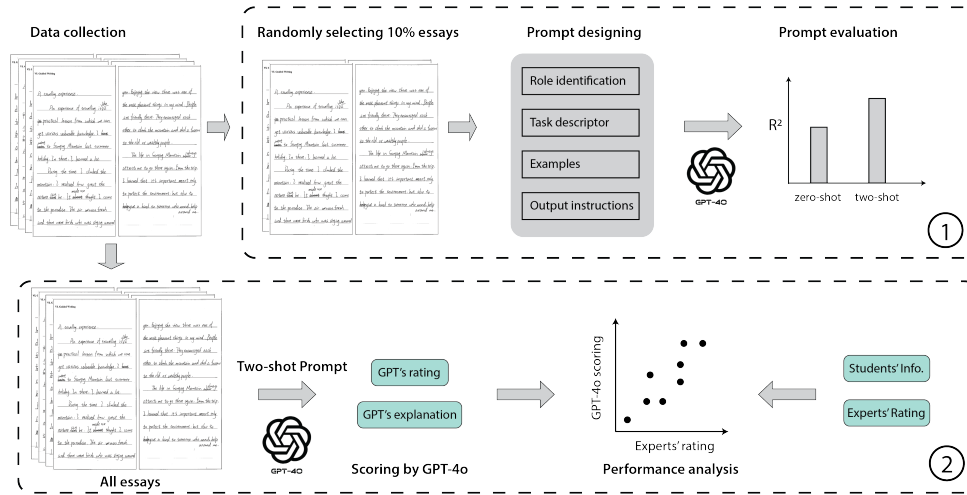


Figure 2: The application of GPT-4o in composition scoring based on image data: research process from prompt strategy design to performance evaluation.

addition, it adopts the Reinforcement Learning from Human Feedback (RLHF) technique, which enables it to better understand and generate content that conforms to human logic. All these provide strong support for the possibility of AES. Due to its rich knowledge reserve, it can comprehensively analyze compositions from multiple aspects such as grammar, vocabulary, logic, and different topics. The capabilities brought about by training on a large amount of data and advanced technologies allow it to be well used for essay scoring in this research without the need for further fine-tuning.

2.3 The Scoring Framework Based on GPT-4o

In this study, we divided the candidate composition image data into two parts. 10% of the samples were used to design and evaluate the prompting strategies of GPT-4o, and the appropriate strategy was determined by comparing the performance of this part of the data under zero-shot and two-shot conditions. After the prompting language test recording is determined, all image data will be used for subsequent large - scale evaluation. The specific process is shown in Figure 2.

2.3.1 The Design and Evaluation of GPT-4o Prompt Strategies Based on Partial Samples. In the module of the design and evaluation of GPT-4o prompt strategies based on partial samples as shown in Figure 2①, this study first randomly selects 10% of the samples from the examinee composition image dataset, and then focuses on the design of prompts for GPT-4o. Finally, by comparing the consistency performance of GPT-4o with the scores of the marking experts under zero-shot and two-shot conditions, these prompt strategies are evaluated. The purpose is to determine the appropriate prompt strategies and lay a solid foundation for the evaluation of subsequent large - scale samples, ensuring the scientificity and accuracy of the evaluation.

2.3.2 The Design and Evaluation of GPT-4o Prompt Strategies Based on Partial Samples. In the comprehensive evaluation module of the scoring performance of GPT-4o under two-shot prompts as shown

in Figure 2②, this study uses the two-shot prompt strategy that performed better in the first part of the evaluation for the remaining 90% of the examinee composition image data and asks GPT-4o to score. Moreover, it further requires GPT-4o to give an explanation for the score, and then combines the scores of the marking experts for the same articles to comprehensively evaluate the performance of GPT-4o in the scoring task through consistency analysis.

2.4 The Design of Prompt

Prompt refers to specific instructions or query information provided to a large language model to guide its behavior and generate the required text information^[14]. In order to generate content that meets the requirements and expectations, it is necessary to comprehensively and accurately describe the requirements in the prompt words^[15]. The effective design and implementation of ChatGPT prompts require in-depth domain expertise and technical proficiency, as these prompts depend on specific threshold criteria. Specifically, role-based prompts can significantly enhance the response quality of large language models^[16]. In view of this, this study defines the role of GPT-4o as an "experienced high school English teacher in China" and designs prompts based on this. Elaborating in detail on the scoring criteria that GPT-4o needs, including specific content and requirements, can make it clear about the scope and focus of the scoring task. Secondly, specifying the output format and requirements of GPT-4o in the prompts can ensure that its output conforms to the expected form and norms of the study. These designs lay the foundation for subsequent research. In addition, in order to clarify which prompt scheme can effectively improve the feasibility and reliability of GPT-4o in handling the scoring of senior one English compositions, this study adopts two strategies as shown in Figure 3 zero-shot prompt and two-shot prompt. The zero-shot prompt in Figure 3A means that when no writing samples scored by marking experts are provided to GPT-4o, it is made to learn only through a specific scoring task or question. The two-shot prompt in Figure 3B is to provide two instances related to the

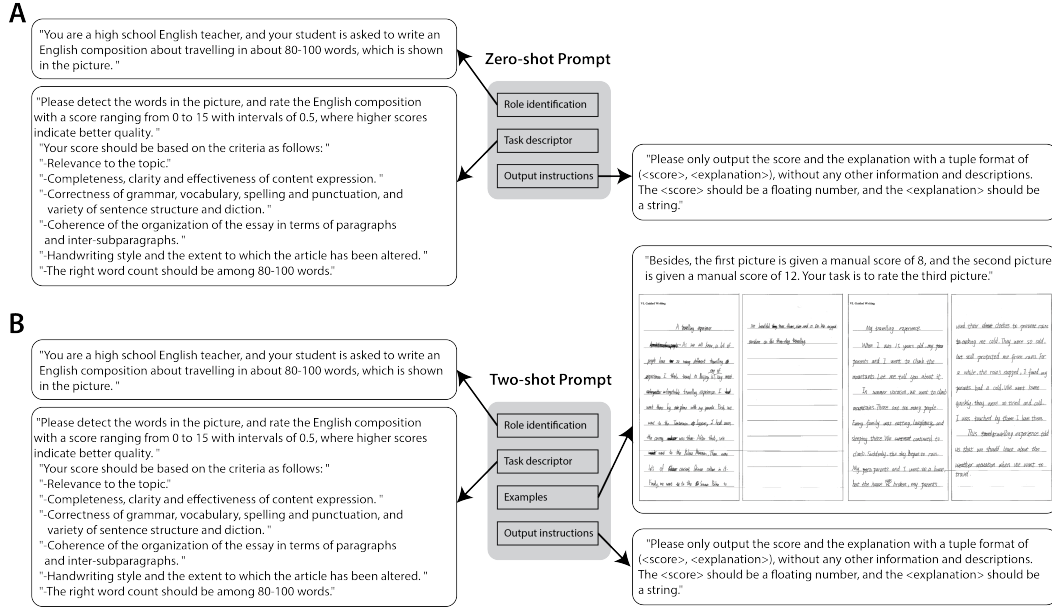


Figure 3: Prompt design of GPT-4o for composition scoring based on image data: zero-shot vs. two-shot.

scoring task and scored by marking experts to the large language model to help GPT-4o understand the scoring rules and methods. In this study, as shown in Figure 3B, we additionally provided two examinee composition images in the prompts and presented the specific scores given by the marking experts. In this study, 110 scoring samples of marking experts were randomly provided to GPT-4o, with the aim of making GPT-4o learn the scoring standards and patterns of marking experts, so as to better simulate the scoring behavior of experts and improve its accuracy and reliability in the task of scoring senior one English compositions.

2.5 Statistical Analysis

We use the GPT-4o API provided by OpenAI to automatically score the image data of examinees' compositions in the Python 3.7 environment. Previous studies have mostly evaluated the scoring accuracy of AES by calculating the consistency between AES scores and human scores^[13]. Therefore, this study adopts the Exact Agreement Coefficient (C), Pearson correlation coefficient, Root Mean Square Error (RMSE), and the probability of GPT-4o scores falling within the confidence interval within different error ranges (P) to verify the performance of GPT-4o in scoring composition images.

2.5.1 Exact Agreement Coefficient. This coefficient measures the degree of exact agreement between two variables. It can reflect the proportion or degree of the situation where the scores of GPT-4o are completely the same as those of the marking experts. The calculation formula is:

$$C = \frac{\sum_{i=1}^n I(x_i = y_i)}{n} \quad (1)$$

The value range of C is usually from 0 to 1. When $C = 1$, it means that the scores of GPT-4o are completely identical to the scores of the marking experts in all composition image scoring. When $C = 0$,

it indicates that the scores of GPT-4o are completely different from the scores of the marking experts in all composition image scoring.

2.5.2 Pearson correlation coefficient. This coefficient can measure the strength and direction of the linear relationship between two variables. It can reflect the degree of linear correlation between GPT-4o scores and the scores of marking experts. The calculation formula is:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2)$$

The value range of r is usually between -1 and 1. When $r = 1$, it indicates that there is a perfect positive linear correlation between the scores of GPT-4o for composition images and the scores of the marking experts; when $r = -1$, it means that there is a perfect negative linear correlation between the two; when $r = 0$, it represents that there is no linear correlation between the two.

2.5.3 Coefficient of determination. This statistic can measure the degree of fit between the model's prediction results and the actual observed values. It can reflect the matching level between GPT-4o scores and the scores of marking experts. The calculation formula is:

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (3)$$

The value range of r^2 is usually between 0 and 1. When $r^2 = 1$, it indicates that the scores of GPT-4o for composition images completely match the scores of the marking experts; when $r^2 = 0$, it means that the scores of composition images have no correlation with the scores of the marking experts, which is equivalent to the result of random guessing. The larger the value of r^2 , the higher the scoring accuracy of GPT-4o for composition images.

2.5.4 Root Mean Square Error. This coefficient is a statistical indicator used to measure the difference between the predicted value and the true value. The scores of the marking experts can be regarded as the true value, and the scores of GPT-4o as the predicted value. Through calculation, the average error degree between the scores of GPT-4o and the scores of the marking experts can be understood. It can reflect the degree of consistency between the scores of GPT-4o and the scores of the marking experts. The calculation formula is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

The value range of $RMSE$ is $[0, +\infty)$. When $RMSE = 0$, it indicates that the scores of GPT-4o in all composition image scoring are completely consistent with those of the marking experts. The larger the value of $RMSE$, the greater the degree of deviation between the scores of GPT-4o for composition images and the scores of the marking experts.

2.5.5 Probability of Falling within the Confidence Interval. This probability measures the likelihood that the GPT-4o score falls within the confidence interval within a given error range e . It can reflect the proportion or degree of falling within the corresponding confidence interval when the error between the GPT-4o score and the mean of the human scores is within a specific range under a given confidence level. This probability helps to evaluate the accuracy and reliability of the GPT-4o score and its closeness to the human score, thus providing an important basis for judging the performance of GPT-4o in specific tasks. For a given error range, the formula for calculating the probability of the GPT-4o score falling within the confidence interval is:

$$P = P(\mu - e \leq x \leq \mu + e) = \int_{\mu-e}^{\mu+e} \frac{1}{\sqrt{2\pi} \frac{\sigma}{\sqrt{n}}} e^{-\frac{(x-\mu)^2}{2(\frac{\sigma}{\sqrt{n}})^2}} dx \quad (5)$$

3 Results

3.1 The Effects of Different Prompts

To understand under which type of prompt GPT-4o has the optimal performance in automatic scoring, this study compared the zero-shot prompt scheme and the two-shot prompt scheme. Each scheme scored 110 composition images and compared the results of LLMs and human scoring one by one. Figures 4A-B show the relationship between GPT-4o scores and the scores of marking experts under different prompting strategies. In these two scatter plots, each small dot represents the scores given by the marking experts and GPT-4o for the composition images. The position of the small dot on the horizontal axis shows the score given by the marking experts for the composition image, and the position on the vertical axis shows the score given by GPT-4o for the same composition image. The small dots have different shades of color, which represents the density of data points. On the graph, the darker small dots mean that the data in this area are more concentrated. Specifically, there are many small dots gathered in this area. Under the ideal state, GPT-4o scores will be completely consistent with the scores of marking experts, that is, the data points completely fall on the gray dotted line of $y = x$. Compared with the zero-shot prompt strategy, the data points in

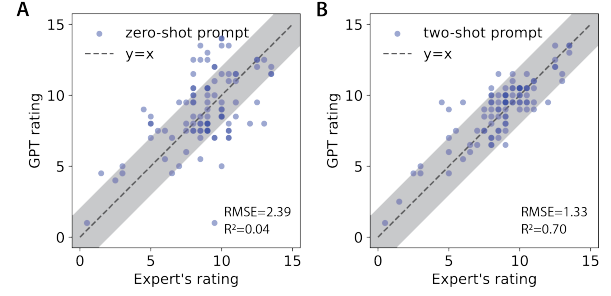


Figure 4: Scatter plots of GPT-4o scores for composition images and scores of marking experts under zero-shot prompts and two-shot prompts.

Table 1: Consistency coefficients between GPT-4o and marking experts' scores under zero-shot and two-shot prompts.

Prompt	C	R	R ²	RMSE	P
Zero-shot prompt	0.07	0.55	0.04	2.39	0.66
Two-shot prompt	0.18	0.85	0.70	1.33	0.93

the two-shot prompt strategy are closer to the dotted line of $y = x$ and the distribution is relatively more concentrated. This indicates that under the two-shot prompt, the consistency error between GPT-4o scores and expert scores is smaller and the goodness of fit is higher.

Table 1 presents the comparison of zero-shot prompt and two-shot prompt in multiple key indicators. Among them, the P-value calculates the probability of falling within the 95% confidence interval with an error range e of ± 2 . After an intuitive comparison of the consistency of the two with the scores of the marking experts from different perspectives, it is found that these indicators all show that the two-shot prompts have a high degree of fit with the expert scores and a small consistency error.

3.2 Descriptive Statistics of GPT-4o Scoring Results

From the data of GPT-4o's scoring results, its average score for the composition image data is 9.34 points, while the average score of the marking experts is 9.21 points. This means that the overall evaluations given by GPT-4o are relatively close to those of the experts.

In terms of the median, the median of GPT-4o's scores is 9.5 points, and the median of the marking experts' scores is 9 points. This reflects that the two kinds of scores are generally the same in terms of the central tendency of distribution.

From the perspective of standard deviation, the standard deviation of GPT-4o's scores is 2.26, and the standard deviation of the marking experts' scores is 2.34. It can be seen that the fluctuation of GPT-4o in the scoring process is slightly smaller than that of the experts, and its scoring consistency is stronger.

Overall, the difference is very small and almost negligible. Whether it is the average score, median, or standard deviation, the scoring results of GPT-4o and the marking experts are very

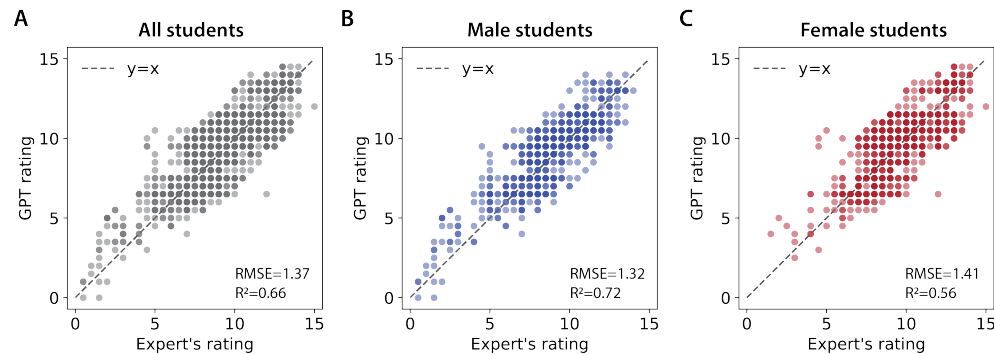


Figure 5: Scatter plots of GPT-4o’s scores for composition images of different groups and the scores of marking experts.

Table 2: Consistency coefficients between GPT-4o and marking experts’ scores for different genders.

Group	C	R	R ²	RMSE	P
All students	0.16	0.82	0.66	1.37	0.90
Male students	0.14	0.86	0.72	1.32	0.91
Female students	0.18	0.78	0.56	1.41	0.88

similar, which fully indicates that GPT-4o has surprisingly accurate and reliable performance in the field of composition scoring.

This result also implies that under certain conditions, GPT-4o can be used as an effective auxiliary tool to help handle large - scale composition scoring work. It can quickly give scores that are almost the same as those of the experts, and in situations where time is tight and tasks are heavy, it can greatly reduce the workload of the marking experts and improve scoring efficiency.

3.3 The Consistency between GPT-4o and Marking Experts’ Scores

Figures 5A-C respectively show the relationship between GPT-4o’s scores for the composition images of all examinees, male examinees, and female examinees and the scores of the marking experts. We can see that when GPT-4o scores the composition images of all examinees, male examinees, and female examinees, there is a certain positive correlation with the score results of the marking experts.

Table 2 presents the comparison of all examinees and examinee groups of different genders in multiple key indicators. After an intuitive comparison of the consistency between GPT-4o and the marking experts’ scores from different perspectives, it is found that under the two-shot prompt, although the accuracy of GPT-4o’s scores for composition images compared with the experts’ scores is relatively low, the correlation is relatively high and the consistency error is small.

In particular, GPT-4o gives higher scores for the composition images of the male group than those of the female group. The possible reason is that the scores given by the marking experts to the male students are relatively dispersed, with a certain distribution density in each score segment, showing a certain normal distribution pattern. However, the score distribution of female students is relatively concentrated, especially with a high density in the high-score segment of 8 - 14 points. If the scoring mechanism of

GPT-4o is more inclined to samples that conform to the normal distribution, then its judgment of the scores of male students may be more accurate. Because in statistical analysis, data with a normal distribution are more easily "understood" and processed by the model. When the score distribution of male students is closer to the normal distribution, GPT-4o can more easily find patterns similar to the training data during scoring, thereby making the consistency between its scores and the experts’ scores higher.

3.4 GPT-4o’s Evaluation and Analysis of Compositions

Figure 6 presents the scoring situation of GPT-4o for composition images in high-score, middle-score, and low-score segments, as well as detailed evaluations of each composition. When GPT-4o scores the composition images of examinees, it can comprehensively consider multiple dimensions of the composition, including content relevance, grammatical accuracy, coherence, writing standardization, and word count requirements. More importantly, GPT-4o can provide detailed evaluation feedback. This feedback is not just a simple score, but goes deep into each component of the examinee’s composition, providing constructive guidance for the examinee. This detailed and targeted feedback mechanism helps examinees deeply understand the advantages and disadvantages of their own writing, and then make targeted improvements and enhancements. To some extent, this is more educationally valuable than simple expert scoring and can more effectively promote students’ learning and growth.

4 Conclusion

This study takes English writing images as the research object and preliminarily explores the application of the multimodal large - language model GPT-4o in examination evaluation. The results show that the performance of GPT-4o in terms of scoring consistency

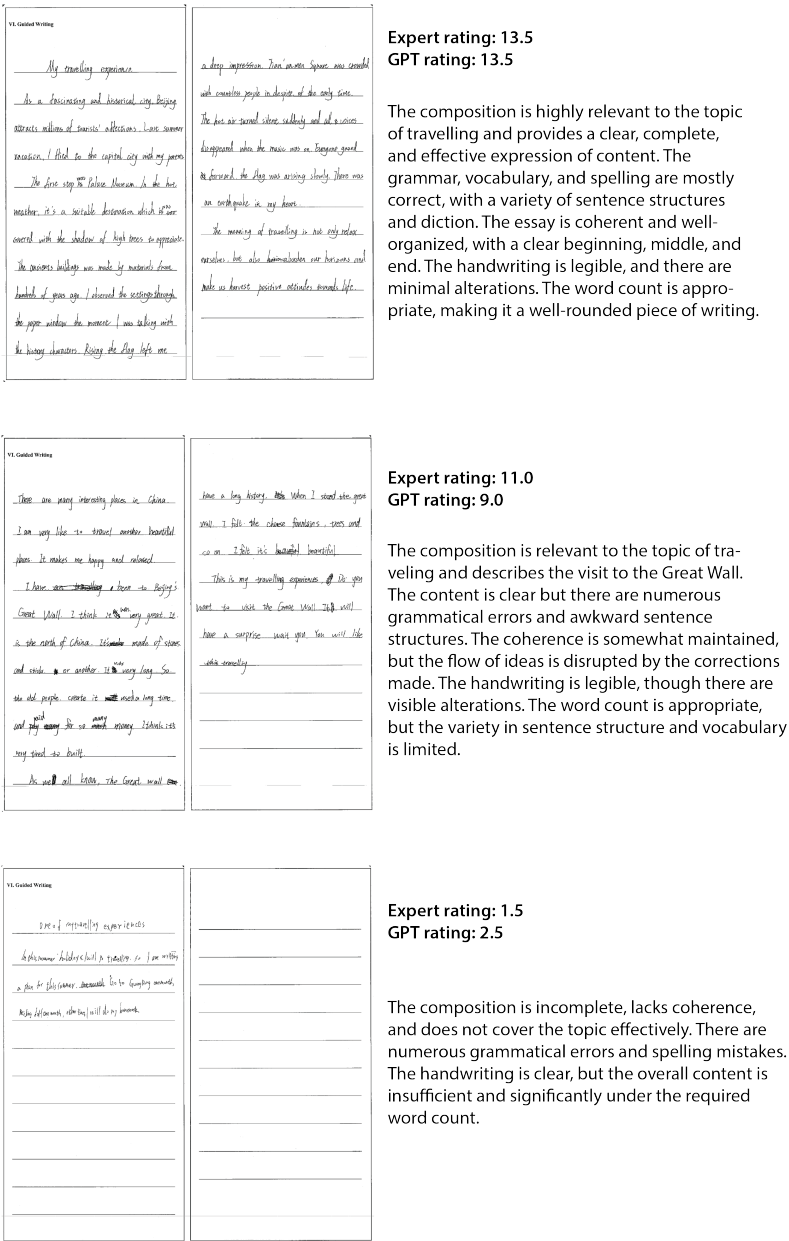


Figure 6: GPT-4o’s Scoring and Evaluation Analysis of Composition Images of Examinees in Different Score Segments.

is within an acceptable range. Compared with existing automatic scoring systems and marking experts’ scores, multimodal LLMs have the following advantages when applied to English writing scoring and feedback.

First, they have outstanding small - sample learning ability. This study finds that multimodal large - language models have the ability to achieve rapid and efficient learning and adaptation with a small number of samples. This excellent feature enables the model to provide high - quality scores and feedback to examinees in a short time. In practical applications, this small - sample - based learning method greatly saves the time and resources required for

model training while ensuring the accuracy and reliability of the evaluation results.

Second, they are easy to operate and have significant cost advantages. Compared with existing commercial automatic scoring systems in China, multimodal large - language models have lower requirements for users’ professional skills. They do not require complex operation processes and professional knowledge reserves, and users can easily get started and quickly apply them to actual examination evaluation work. This not only reduces labor costs

but also reduces operational errors that may be caused by insufficient professional skills, making the examination evaluation process smoother and more efficient.

Third, multimodal analysis facilitates personalized feedback. Multimodal LLMs can deeply analyze the content of images and then provide targeted and personalized evaluation feedback for each examinee. This personalized feedback mechanism is of great significance for the improvement of examinees' writing skills and thinking abilities. By accurately pointing out the advantages and disadvantages of examinees during the writing process and giving specific improvement suggestions, examinees can be more clear about their learning directions, conduct targeted training and improvement, and effectively improve their writing ability and thinking level.

This study takes the scoring of English composition images as the starting point. In the future, by further optimizing algorithms and training data, the degree of fit with expert scoring can be made higher. Even in some details, it can provide more valuable references and ideas for marking experts, jointly promoting the development and improvement of the composition scoring system.

Moreover, the application scenarios of LLMs in examination evaluation are far more than this. In future research, its application range can be further expanded to explore its application in the evaluation of open-ended questions in other subjects. For example, in the solutions of open-ended application problems in mathematics, the experimental design and analysis in physics, and the evaluation of discussion questions in history, multimodal large-language models may all play important roles.

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